

CUSTOMER SUPPORT TICKET PRIORITY
PREDICTION
AND AUTOMATED ASSIGNMENT SYSTEM
USING AGENTFORCE

VIVEKANANDHA COLLEGE OF TECHNOLOGY
FOR WOMEN [6130]

DEEPIKA V (613023104015)	TEAM LEADER	deepikapapitha@gmail.com
LAVANYA B (613023104055)	TEAM MEMBER	lavanyababu2005@gmail.com
MANISHA K (613023104060)	TEAM MEMBER	manishakrishnan2005@gmail.com
GOPIKA R (613023104027)	TEAM MEMBER	gopikaramesh63@gmail.com

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

User Story:

The Customer Support Ticket Priority Prediction and Automated Assignment System combines Salesforce and Agentforce to read the description of each customer support ticket, rate its urgency as High, Medium, or Low, and start the right backend automation. It relies on the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent called Support Ticket Priority Analysis. Tickets rated High generate an urgent handling task and go to a senior support agent, which cuts manual work and shortens response times. The goal is to remove repetitive triage work and give customers faster answers.

1. INTRODUCTION

1.1 Project Overview :

Support teams handle a high volume of tickets every day, and sorting and routing them by hand is slow. This project delivers a Salesforce-based solution that reviews ticket details, judges how urgent each case is, selects the suitable support level, and launches automation for critical issues.

1.2 Purpose :

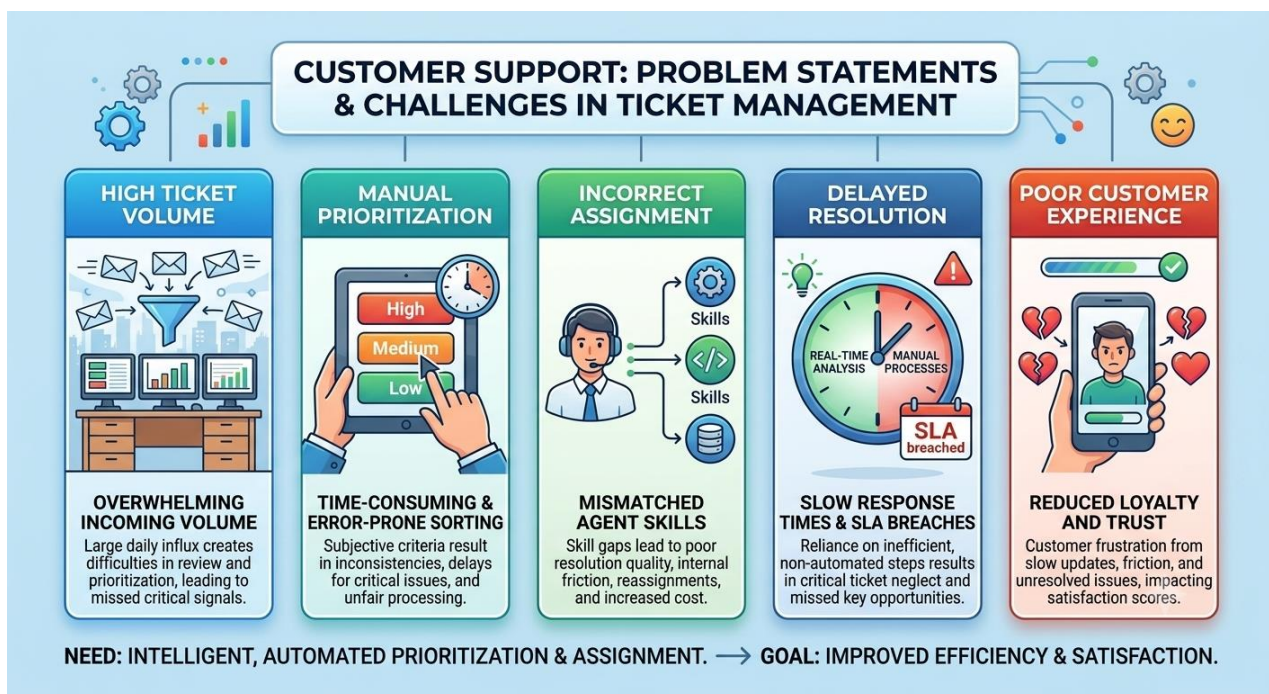
- Rank incoming support tickets as High, Medium, or Low priority automatically.
- Fetch the most recent support ticket linked to a customer Account.
- Cut down the manual work of prioritizing and assigning tickets.
- Generate an urgent handling task whenever a ticket is High priority.
- Let Agentforce review ticket details and start the backend automation.
- Speed up ticket response and resolution.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	handle the most urgent tickets first	new tickets are rarely ranked by urgency automatically	ranking is done by hand	annoyed
PS-2	Customer	see urgent problems solved promptly	serious problems can sit behind less important tickets	support teams are handling heavy ticket loads	worried
PS-3	Support Manager	route tickets to the correct support tier	routing often needs a manual decision	urgency is not judged the same way every time	swamped

PS-4	Support Agent	spot urgent problems in the description text	key words are easy to overlook	descriptions carry a wide range of urgency cues	unsure
PS-5	Support Manager	limit the risk of missing SLAs	aging open tickets can go unseen	SLA risk requires extra review	worried



2.2 Empathy Map Canvas

User Persona: Support Agent

Think & Feel: Needs critical issues flagged right away so that urgent customer cases are seen first.

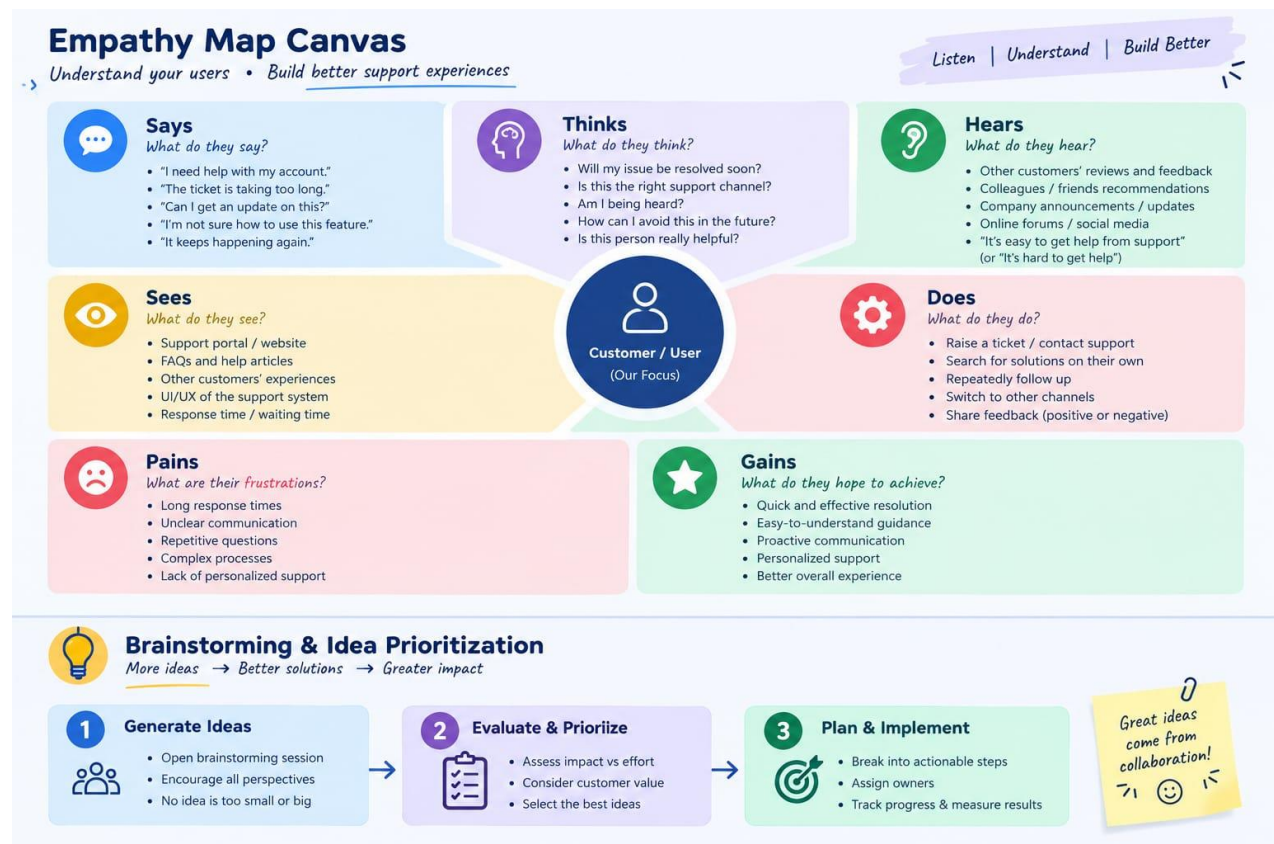
Hear: Takes in customer issue reports and support requests.

See: Ticket status, issue type, description, priority, assignment, and SLA risk indicators.

Say & Do: Goes through tickets, works assigned cases, and concentrates on urgent issues.

Pain: Prioritizing by hand, overlooked urgency cues, and slow handling of critical tickets.

Gain: Priority set automatically, clearer assignment, urgent tasks created, and less manual work.



2.3 Brainstorming & Idea Prioritization

Ideas Considered

- Handwritten ticket registers
- Tracking tickets in spreadsheets
- A separate standalone support portal
- Salesforce CRM with Agentforce

Final Selection

Salesforce CRM selected due to:

- Workflow automation using Salesforce Flow
- AI-assisted ticket review with Agentforce
- Automatic classification of ticket priority
- Automatic creation of urgent tasks
- Less manual effort for the support team



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

Ticket Creation → Account Identification → Ticket Retrieval → Description
Analysis → Priority Classification → Agent Assignment → Urgent Task
Creation → Support Handling → Resolution



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Creating and managing support tickets
FR-2	Linking tickets to Accounts and Contacts
FR-3	Reading and analyzing ticket descriptions
FR-4	Automatic assignment of priority levels
FR-5	Task creation for High-priority tickets
FR-6	Assignment to the right support agent
FR-7	Checking tickets for SLA breach risk
FR-8	Conversational analysis through Agentforce

Non-Functional Requirements

NFR	Description
Usability	Agentforce conversations are clear and priority messages are easy to understand
Reliability	Priority results stay consistent because they follow the configured conditions
Performance	Fast lookup and analysis of the most recent support ticket

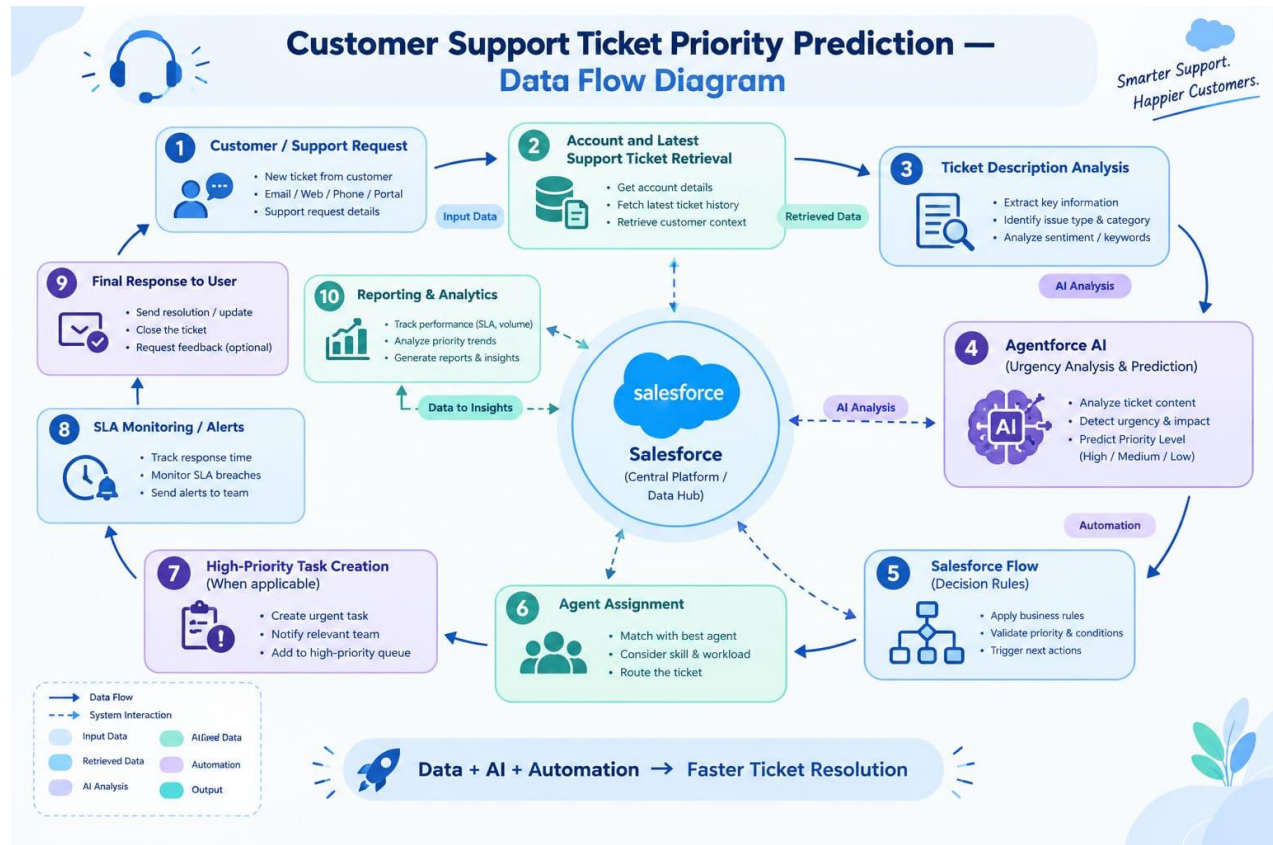
Security	Salesforce access to customer and ticket data is kept under control
Maintainability	Flow and Agentforce settings are easy to edit
Scalability	Salesforce automation handles growing support ticket volumes

3.3 Data Flow Diagram

Flow:

Customer / Support Request → Account → Latest Support Ticket → Ticket Description
→ Analyze Description → Priority Level → High/Medium/Low Decision → Agent
Assignment → High-Priority Task (if applicable) → Final Response

Agentforce: User provides Account Name → Agentforce Support Ticket Priority Analysis
→ Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output →
Response to User.



3.4 Technology Stack

Layer	Technology
Platform	Salesforce
Database	Support_Ticket_Intelligence__c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce
Agent Configuration	Support Ticket Priority Analysis subagent

Automation	Flow Decision, Assignment and Create Records
Output	Response delivered in the Agentforce conversation

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Tickets are ranked by hand	Flow reads the description and sets the priority	Quicker spotting of urgent cases
Urgent issues can be overlooked	Keyword-based high-priority decision	Critical cases receive prompt attention
Tasks created manually	Flow creates a Task for High priority tickets	Less manual effort
Support assignment done by hand	Flow sets the assigned support level	More uniform handling
SLA risk may go unnoticed	Optional SLA risk check	Improved risk visibility
Need for a conversational interface	Agentforce subagent	AI-supported conversations



4.2 Proposed Solution

Functional Requirements

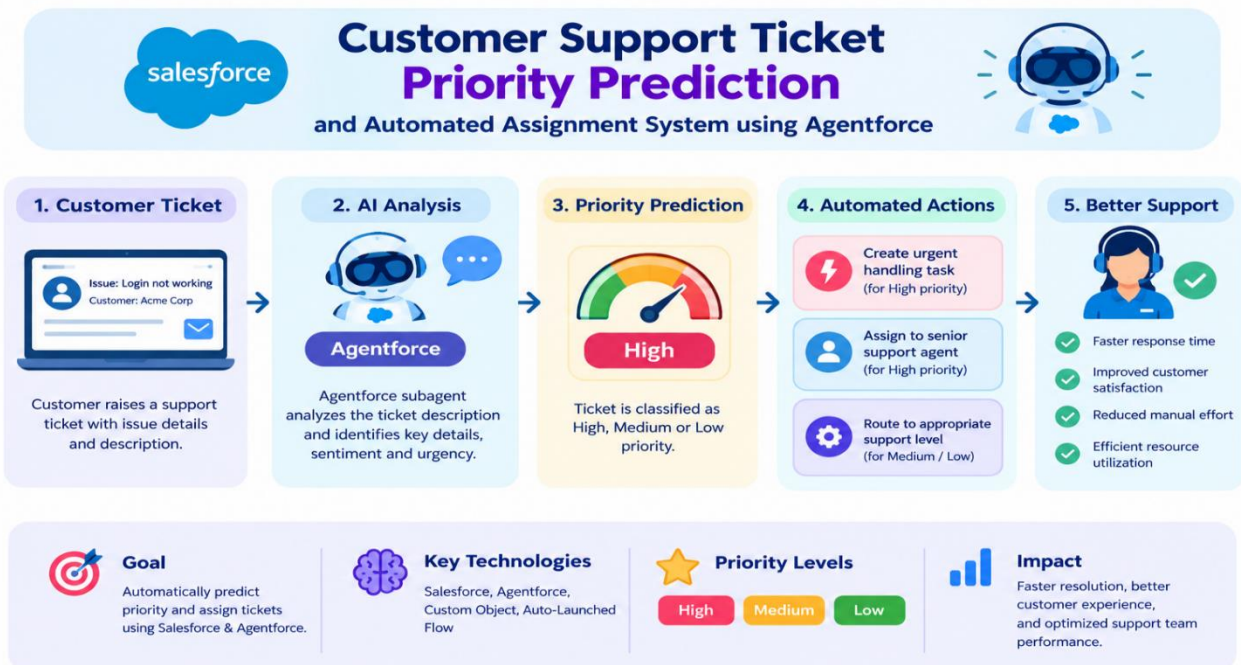
Parameter	Description
Problem	Slow, inconsistent manual prioritization and assignment that delay critical support issues
Solution	A Salesforce solution for Customer Support Ticket Priority Prediction and Automated Assignment
Innovation	Agentforce AI combined with an Auto-Launched Flow to read ticket descriptions, classify their priority, choose the support level, and trigger urgent-task automation

Impact	Critical tickets are spotted sooner, manual work drops, responses are quicker, and support teams are more productive
Business Model	Support operations that involve customers, support agents, support managers, and AI-assisted ticket handling
Scalability	Handles growing ticket volumes through Salesforce automation, and leaves room for additional assignment rules, deeper SLA monitoring, and more priority indicators

4.3 Solution Architecture

Components:

Users / Support Team → Agentforce → Support Ticket Priority Analysis →
Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority
Decision → Task / Assignment → Final



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile delivery approach
- Epics → Stories → Story Points
- Work organized into sprints
- Estimates guided by velocity calculation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to set up the Salesforce environment for Support Ticket Intelligence so that the support automation solution	3	High	Member1
Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			can be built.			
Sprint-2	Data Modeling	USN-2	As an admin, I want to build the Support Ticket Intelligence object with the needed fields and relationships so that customer ticket data is stored in an organized way.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a support user, I want support tickets to be associated with Accounts and Contacts so that the latest customer ticket can be identified for analysis.	3	High	Member3
Sprint-3	Automation	USN-4	As a support agent, I want an Auto-Launched Flow that fetches the latest ticket and examines its description so that priority can be decided automatically.	5	High	Member4

Sprint-3	Automation	USN-5	As a support agent, I want tickets automatically labeled High, Medium, or Low according to set urgency conditions so that critical issues are handled first.	5	High	Member4
Sprint-4	Agentforce & Automation	USN-6	As an AI Agent (Agentforce), I want to review ticket details and run the backend Flow so that manual prioritization is minimized.	8	High	Member1
Sprint-4	Task & Assignment	USN-7	As a support manager, I want High-priority tickets to automatically produce an urgent handling task and be given to the proper support level so that critical tickets are resolved quickly.	5	High	Member2
Sprint-5	SLA Management	USN-8	As a support manager, I want the system to check older unresolved tickets for SLA breach risk so that potential SLA problems are identified.	5	Medium	Member3
Sprint-6	Agentforce & Conversational Support	USN-9	As a support user, I want to give Agentforce an Account Name and get back the ticket ID, priority, assigned agent, and action message so that I can follow the ticket handling result in a conversation.	5	Medium	Member4

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

Project Tracker, Velocity & Burndown Chart: (4 Marks)

6. Project Development Phase:

What you'll learn

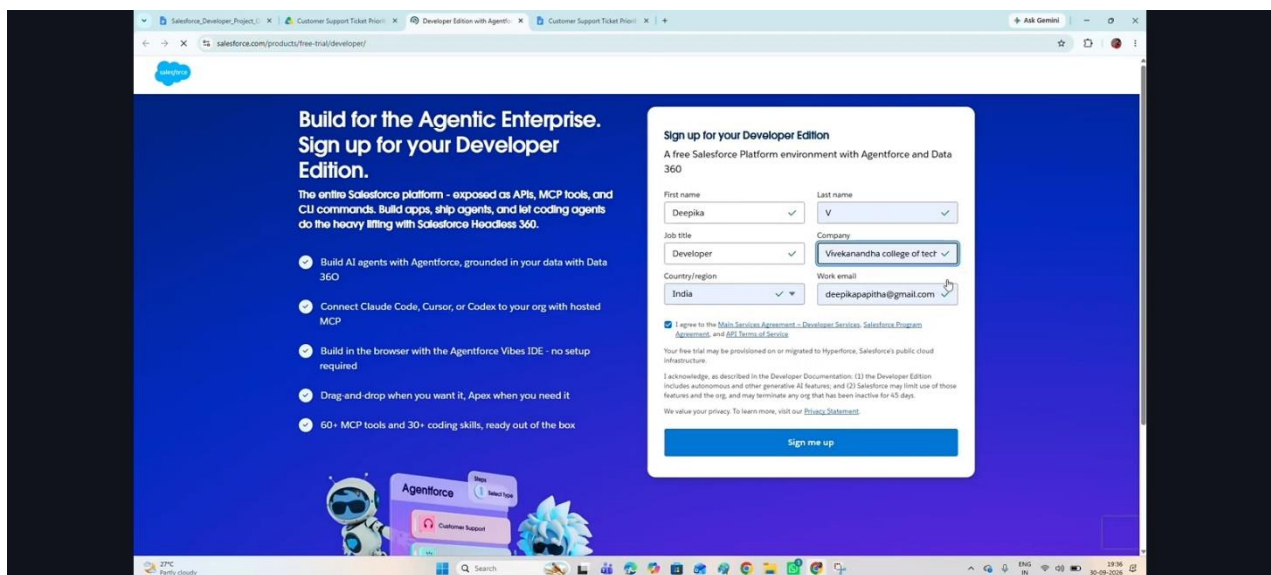
1. Data Modelling
2. Fields and Relationships
3. Lightning App Builder
4. Record Triggered Flows
5. Apex and Apex Triggers
6. Batch Apex
7. Scheduled Apex

Milestone 1: Salesforce Account

Activity 1: Creating Developer Account

Setting up a developer org in Salesforce.

1. Go to <https://developer.salesforce.com/signup>
2. Fill in the sign-up form with these details :



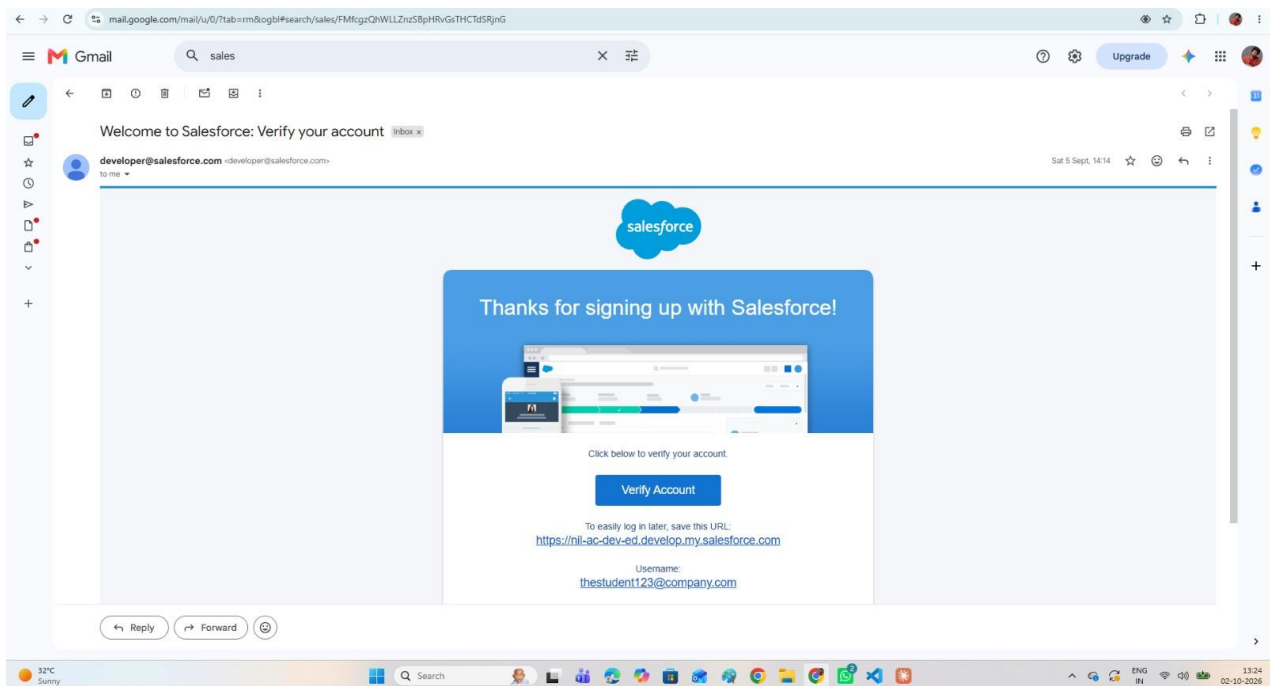
- 1) First name & Last name
- 2) Email
- 3) Role : Developer
- 4) Company : College Name

- 5) County : India
- 6) Postal Code : pin code
- 7) Username : should be combination of your name and company The email does not have to be real; any address in this format works :

username@organization.com Click on sign me up once the form is filled in.

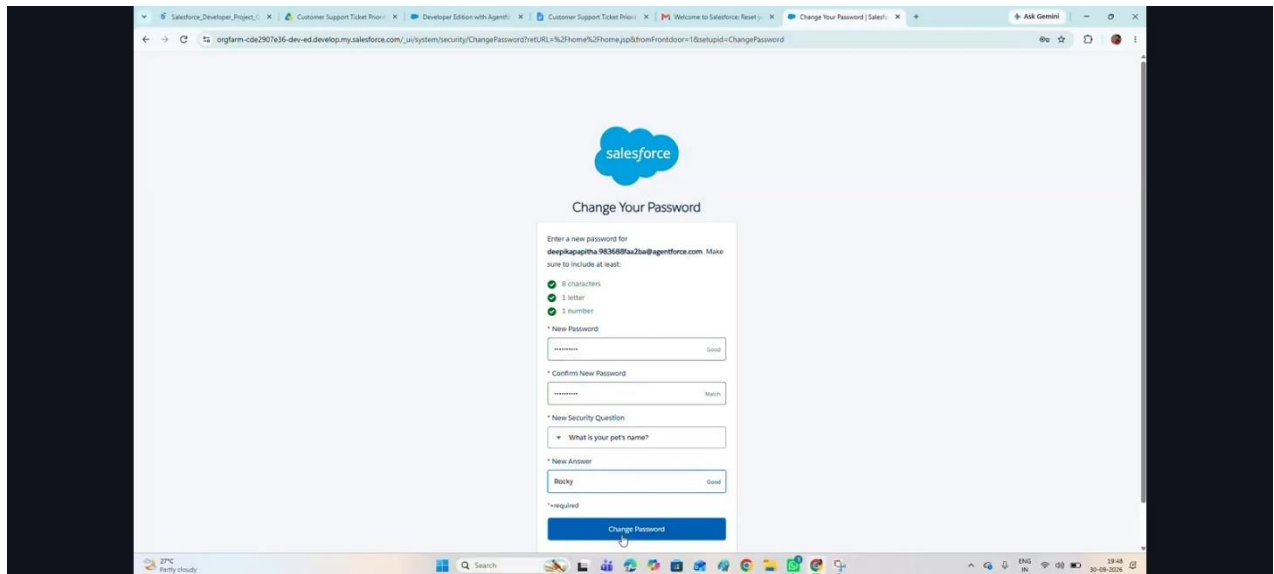
Activity 2: Account Activation

1. Open the inbox of the email address used for sign-up and click on verify account to activate it. The email can take 5-10 mins to arrive.

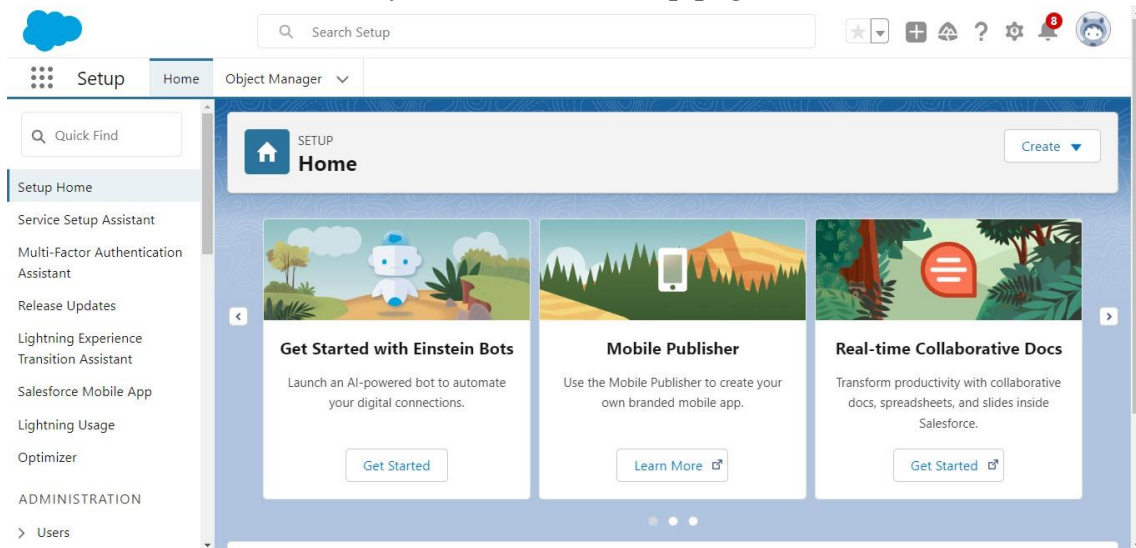


2. Click on Verify Account

3. Set a password, answer a security question, and click on change password



4. You will then be redirected to your Salesforce setup page.



Milestone 2: Data Management-Objects

Custom Object

Object Name:

Support Ticket Intelligence

API Name:

Support_Ticket_Intelligence__c

Fields

Field Label	API Name	Data Type	Description
Ticket Number	Ticket_Number__c	Auto Number	TKT-{0000}
Customer	Customer__c	Lookup (Account)	Linked Account
Contact	Contact__c	Lookup (Contact)	Linked contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Details of the issue
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Date of the ticket
Assigned To	Assigned_To__c	Lookup (User)	Assigned support agent
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	SLA risk indicator
Resolution Time (hrs)	Resolution_Time__c	Number	Time to reslove

Auto-Launched Flow Logic (Modified for this Project)

1. Flow Type

Auto-Launched Flow (No Trigger or called from Agentforce)

2. Variables

Input Variable

Label	API Name	Type	Input
Account Name	varAccountName	Text	Yes

Output Variables

Label	API Name	Type	Output
Account Id	varAccountId	Text	Yes
Ticket Id	varTicketId	Text	Yes
Priority Level	varPriorityLevel	Text	Yes
Assigned Agent	varAssignedTo	Text	Yes

Action Message	varActionMessage	Text	Yes
-------------------	------------------	------	-----

3. Step-by-Step Flow Implementation

Step 1: Get Account

Element: Get Records

- Label: Get Account
- Object: Account
- Condition: Name Equals {!varAccountName}
- Sort: CreatedDate DESC
- Store: Only first record

Step 2: Assignment (Store Account Id)

Element: Assignment

- Label: Store Account Id Set:
- varAccountId = {!Get_Account.Id}

Step 3: Get Ticket Record

Element: Get Records

- Label: Get Ticket
- Object: Support_Ticket_Intelligence__c
- Condition: Customer__c Equals {!varAccountId}
- Sort: CreatedDate DESC
- Store: Only first record

Step 4: Assignment (Store Ticket Id)

Element: Assignment

- Label: Store Ticket Id Set:
- `varTicketId = {!Get_Ticket.Id}`

Step 5: Decision (Analyze Description)

Element: Decision

- Label: Analyze Description

Outcomes:

High Priority

- Description contains:
 - "urgent"
 - "not working"
 - "failure"

(Use multiple conditions with CONTAINS)

Medium Priority

- Description contains:
 - "issue"
 - "slow"
 - "delay" **Default**

Outcome

- Low Priority

Step 6: Assignment (Set Priority Level)

High Path

- `varPriorityLevel = "High"`

Medium Path

- `varPriorityLevel = "Medium"`

Low Path

- `varPriorityLevel = "Low"`

Step 7: Decision (High Priority Check)

Element: Decision

- Label: Is High Priority

Condition:

- `varPriorityLevel Equals "High"`

Step 8: Create Task (Only High Priority)

Element: Create Records

- Object: Task

Set Fields:

- Subject = "Urgent Ticket Handling" • WhatId = `{!varTicketId}`
- Priority = High
- Status = Not Started

Step 9: Assignment (Assign Agent Name)

Element: Assignment

- Label: Set Assigned Agent

Set:

- `varAssignedTo = "Senior Support Agent"`

Step 10: Optional SLA Check

Decision: SLA Risk

Condition example:

- If Created Date older than 2 days Then:
- (Optional internal flag or message)

Step 11: Final Output Message

Element: Assignment

- Label: Set Final Message Set varActionMessage:
- High:
"High priority ticket detected. Assigned to senior agent."
- Medium:
"Ticket marked as medium priority. Will be handled shortly."
- Low:
"Tickets are low priority and queued for processing."

Step 12 : Save Flow

Label Name : Support_Ticket_Intelligence

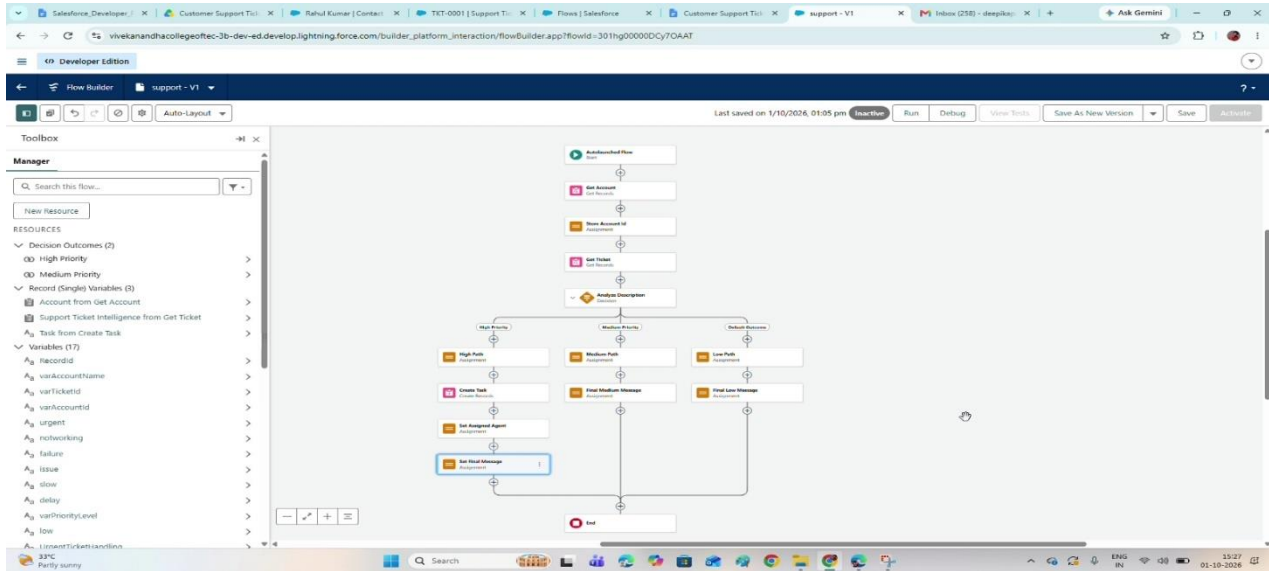
Active Flow

4. Final Flow Structure

Start

- Get Account
- Assignment (Account Id)
- Get Ticket
- Assignment (Ticket Id)
- Decision (Analyze Description)
- Assignment (Priority Level)
- Decision (High Priority)
- Create Task
- Assignment (Assigned Agent)
- Optional SLA Decision
- Assignment (Final Message)
- End

Flow OutPut :



The screenshot shows the 'Support Ticket Intelligence' interface. On the left, there is a list of tickets with columns for 'Customer', 'Contact', 'Issue Type', 'Description', 'Priority Level', 'Status', and 'Created Date'. The ticket TKT-0002 is highlighted. On the right, there is a summary of upcoming tasks. The summary shows two tasks, both labeled 'Urgent Ticket Handling', with a status of 'No due date'. A 'View More' button is located at the bottom of the summary.

5. Final Outcome

- Takes **Account Name as input**
- Fetches latest support ticket
- Analyzes description using keywords
- Sets priority automatically
- Creates task for urgent cases
- Returns output to Agentforce

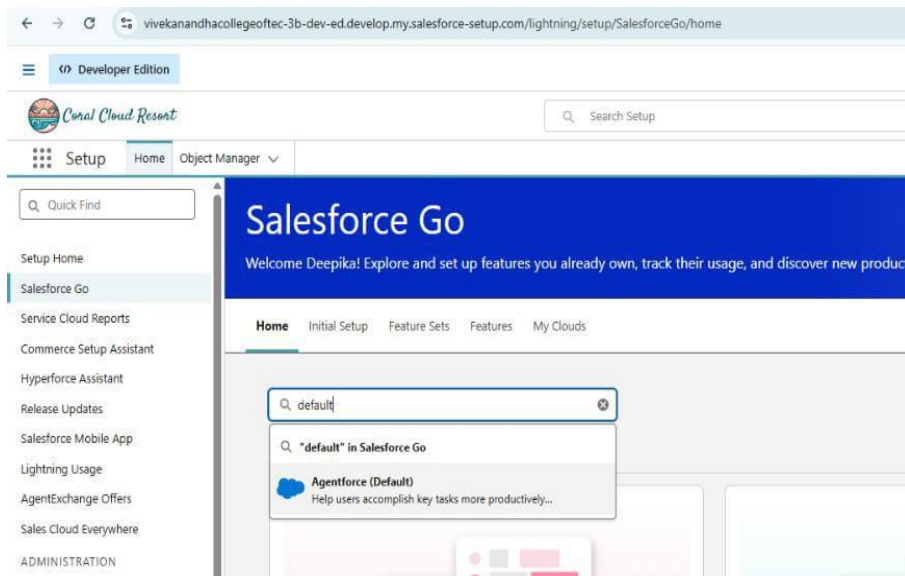
Agentforce Configuration

Steps :

1. Click the gear icon and choose Salesforce Go

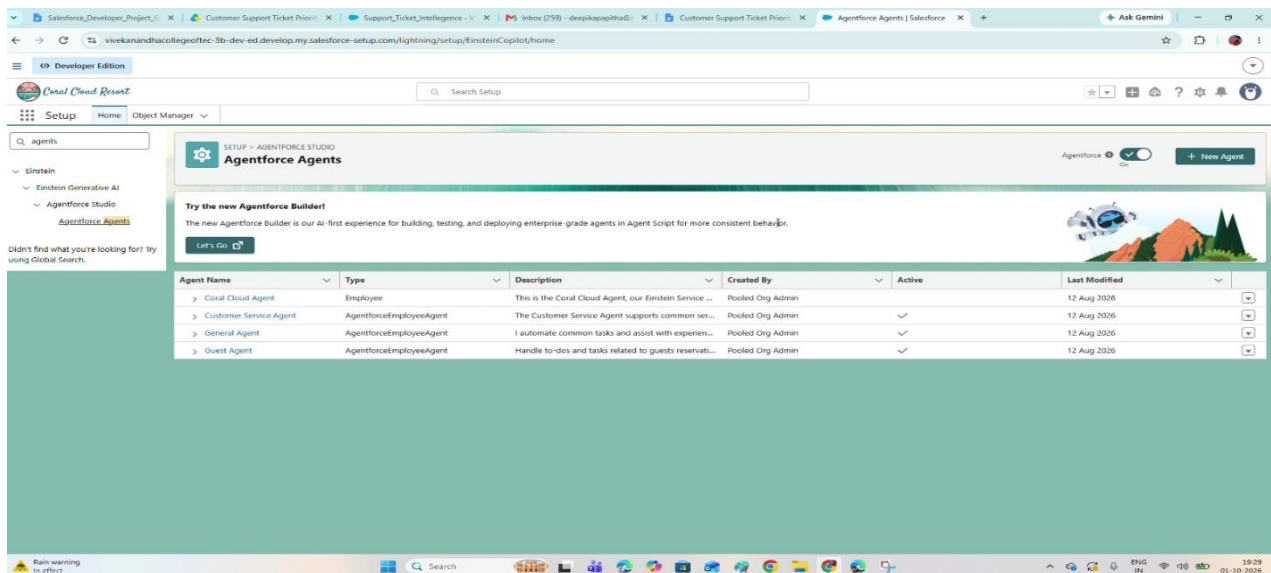


2. Search for Default, pick Agentforce (Default), and enable it



3. Select Get Started ⇒ Turn On ⇒ Confirm

4. Type Agents in the Quick Find box and enable the agent; the Default Agent will then appear



5. Choose the Default Agent and click Open in Builder

SETUP > AGENT DETAILS
Agentforce (Default) Version 1 (Active) **Open in Builder**

API Name	Type	Active
Copilot_for_Salesforce_99045161979297	Employee	☒

Details System Messages Language Settings

6. New to add a new SubAgent

Subagents

Manage the subagents assigned to your agent. To make changes, your agent must be deactivated.

Search subagents... Search subagents...

9 items • Sorted by Subagent Label(asc)

Subagent Label ↑	
Account Management	
Case Management	
Delivery Issues	
Escalation	
General CRM	
General FAQ	
Order Inquiries	
Reservation Management	
Service Customer Verification	

7. Click Next

Put your subagents to the test

Start a conversation to preview how your agent builds a plan and executes actions based on user interactions.

Conversation Preview

Let's chat!

Hi, I'm an AI customer service agent. How can I help?

Describe your task or ask a question...

Create a Subagent

What do you want this subagent to do? (Optional)

Describe the job you want this subagent to perform.

Cancel Next

Agentforce Subagent Configuration

Name

Support Ticket Priority Analysis

API Name

Support_Ticket_Priority_Analysis

Classification Description

This topic reviews customer support tickets and sets priority from the issue description and its urgency. It answers user requests about ticket status, issue severity, and support prioritization.

Scope

Your only task is to analyze support ticket descriptions, decide the priority level (High, Medium, Low), and start the backend automation for task assignment. Do not take on unrelated requests such as billing, subscription management, or account updates.

Instructions

Carry out these steps whenever this topic is triggered:

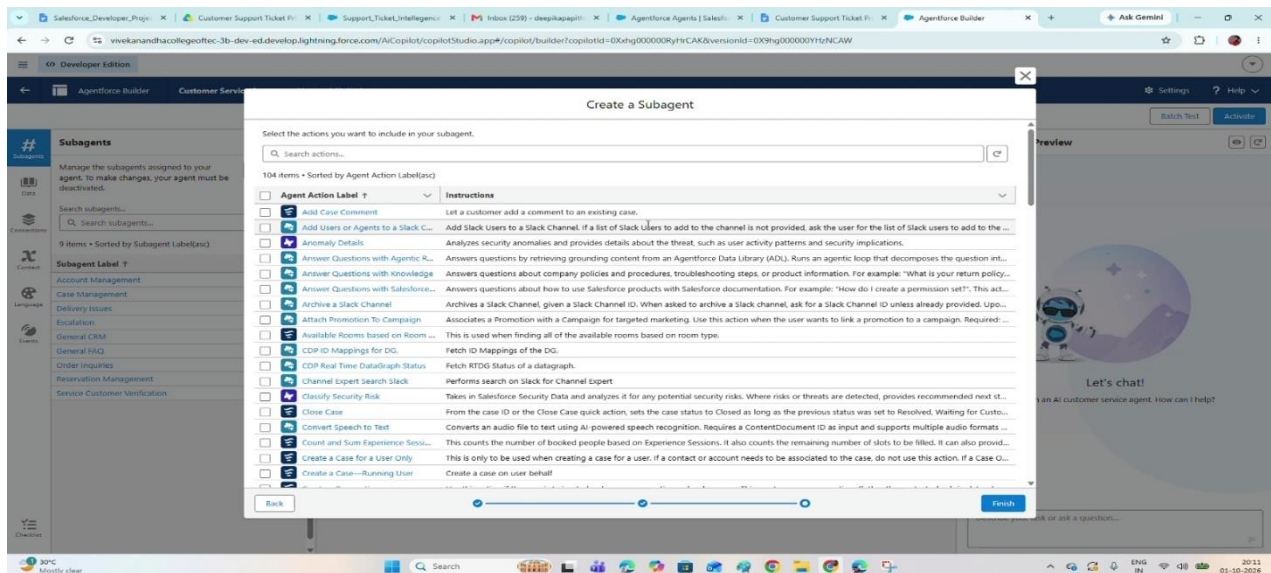
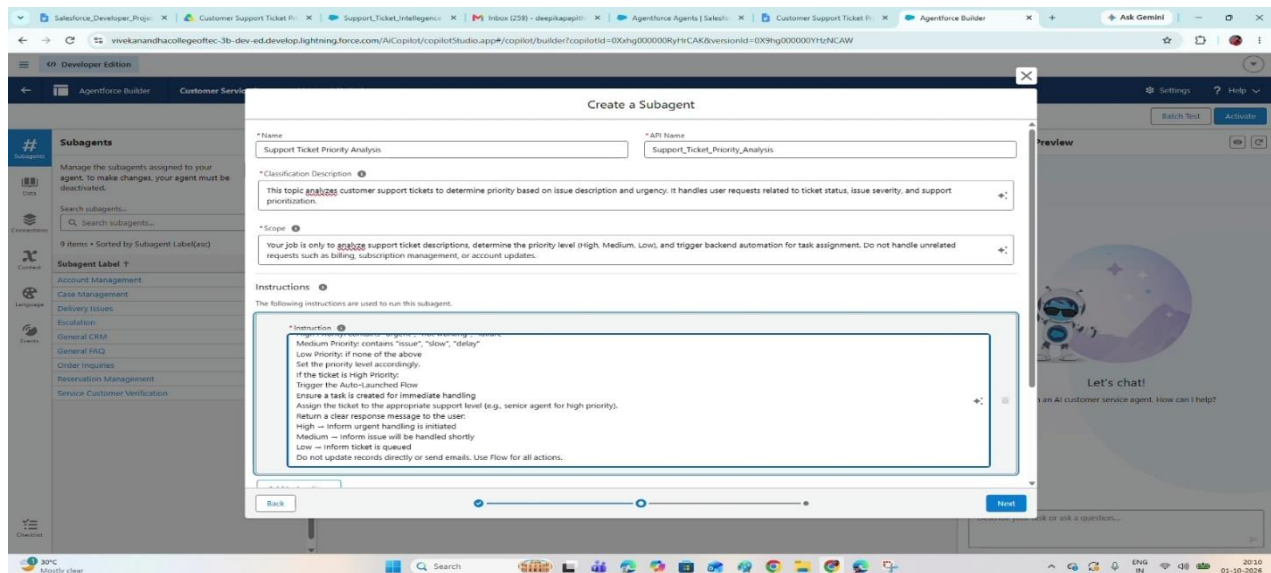
1. Request the Account Name from the user if it has not been provided.
2. Fetch the most recent support ticket linked to that Account.
3. Read through the ticket description and analyze it.
4. Decide the priority using these keywords:
 - High Priority: contains "urgent", "not working", "failure"
 - Medium Priority: contains "issue", "slow", "delay"
 - Low Priority: if none of the above
5. Record the matching priority level.
6. When the ticket is High Priority:
 - Run the Auto-Launched Flow
 - Make sure a task is created for immediate handling

7. Route the ticket to a suitable support level (for example, a senior agent for high priority).

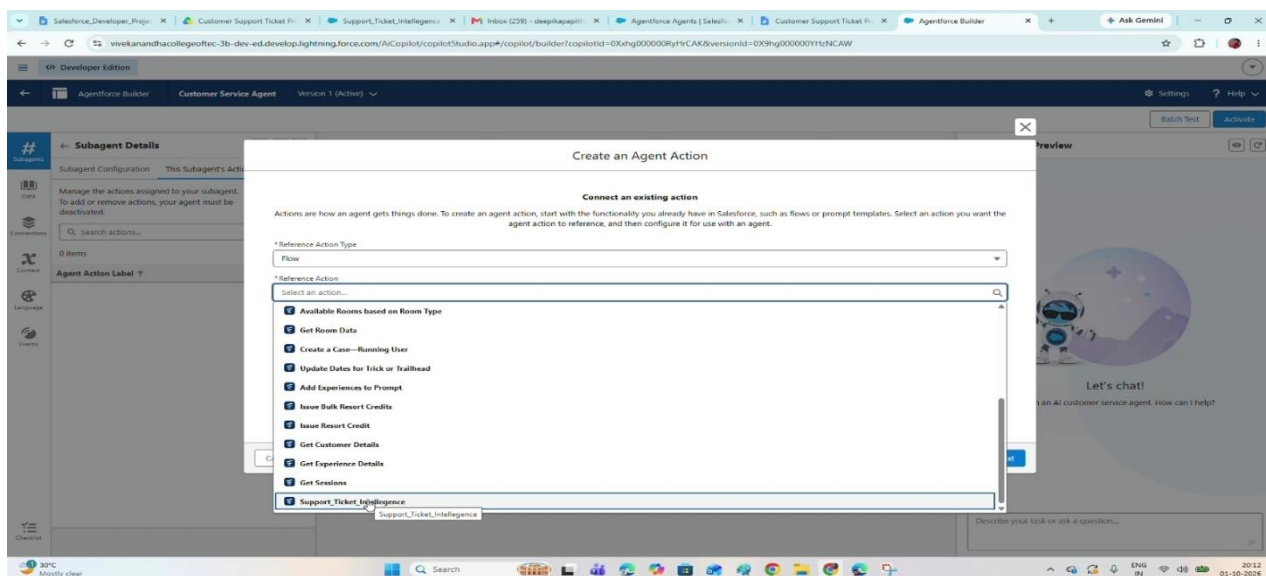
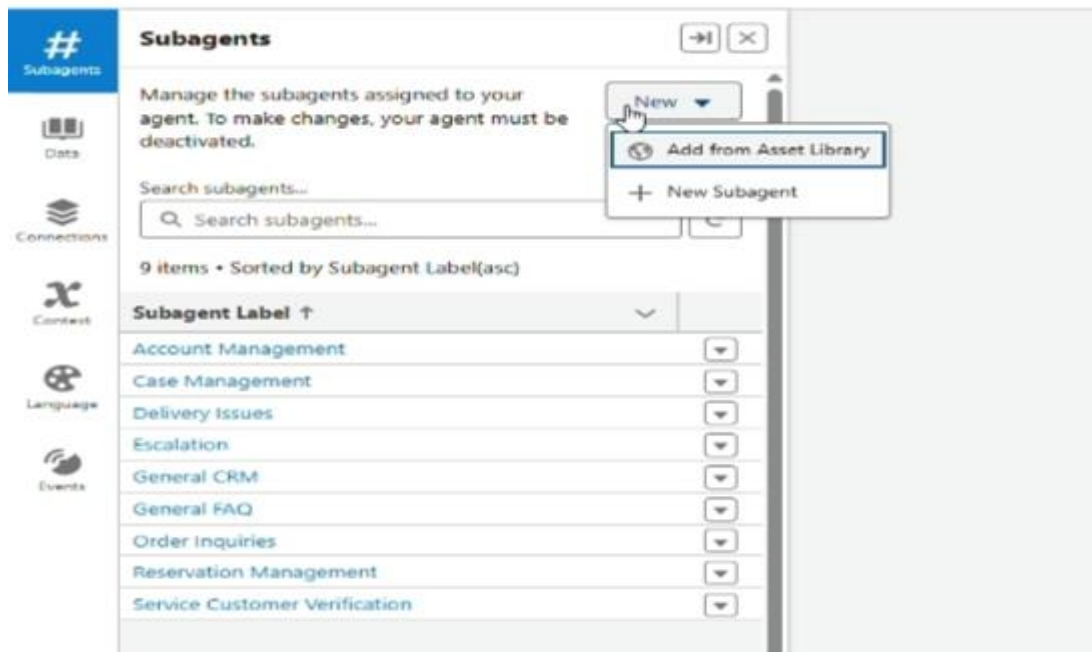
8. Give the user a clear response message:

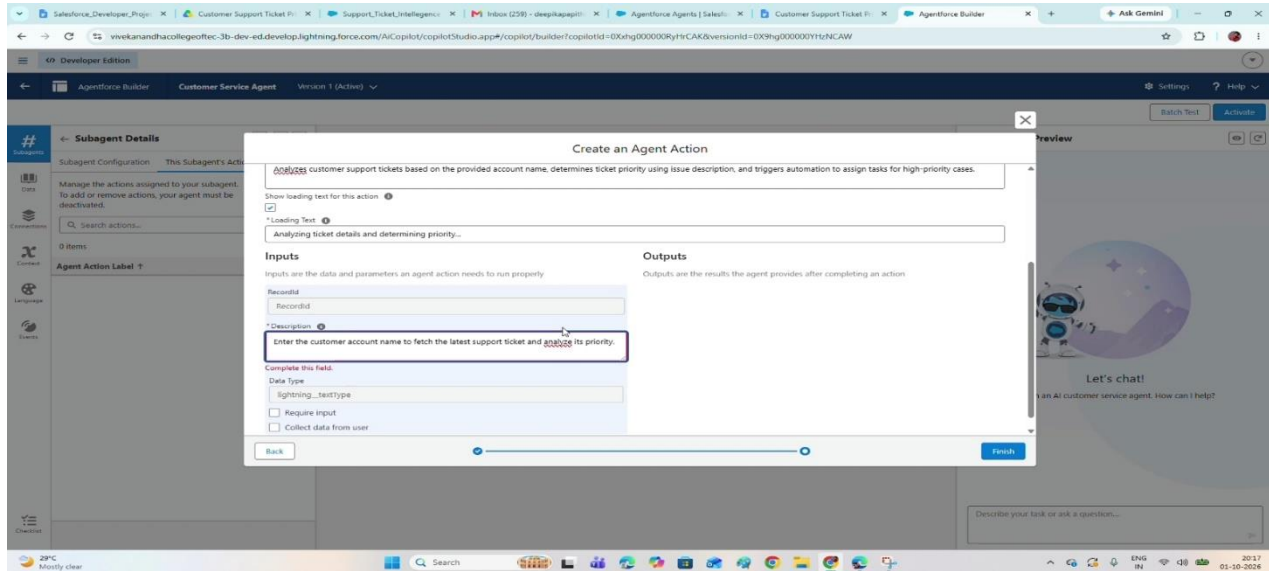
- High → Say that urgent handling has started
- Medium → Say that the issue will be handled soon
- Low → Say that the ticket is queued

9. Never edit records directly or send emails; perform every action through Flow.



Select Topic





Agent Action Description

Reviews customer support tickets for the supplied account name, determines ticket priority from the issue description, and triggers automation that creates tasks for high-priority cases.

Loading Text

Reviewing ticket details and deciding the priority...

Input

varAccountName

- **Description**

Provide the customer account name so the latest support ticket can be fetched and its priority analyzed.

- **Data Type**

lightning__textType

- **Require Input**

Enabled

- **Collect Data From User**

Enabled

Outputs

varAccountId

- **Description**

Gives back the unique ID of the account used to find the support ticket.

- **Data Type**

lightning__textType

- **Show in Conversation**

Enabled

- **Output Rendering**

Text

varActionMessage

- **Description**

Shows the final message with the ticket priority and the actions taken.

- **Data Type**

lightning__textType

- **Show in Conversation**

Enabled

- **Output Rendering**

Text

varAssignedTo

- **Description**

Shows which support agent or team has been assigned to the ticket

- **Data Type**

lightning__textType

- **Show in Conversation**

Enabled

- **Output Rendering**

Text

varPriorityLevel

- **Description**

Holds the ticket priority level (High, Medium, Low) worked out from the issue description.

- **Data Type**

lightning__textType

- **Show in Conversation**

Enabled

- **Output Rendering**

Text

varTicketId

- **Description**

Gives back the unique ID of the support ticket the system analyzed.

- **Data Type**

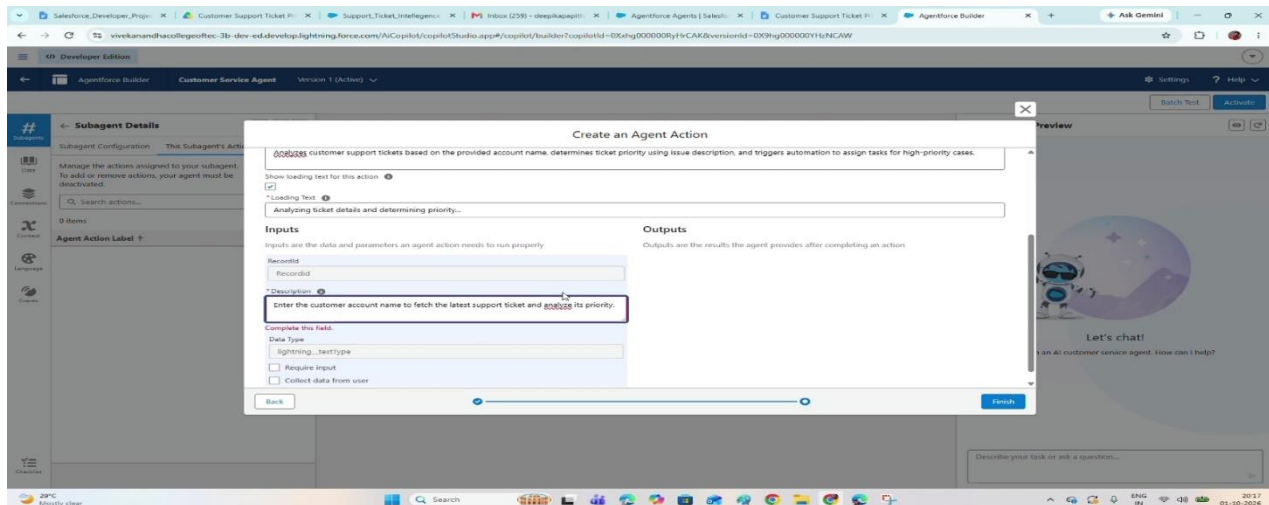
lightning__textType

- **Show in Conversation**

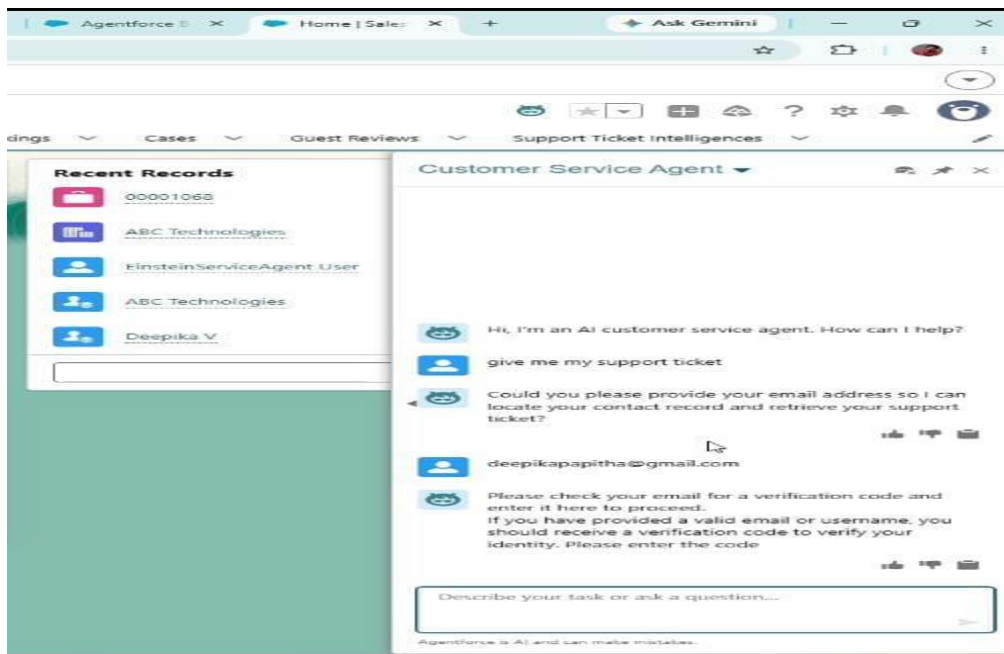
Enabled

- **Output Rendering**

- Text



Final OutPut :



Note : To extend this scenario, add more variables to the flow and map them to the agent. This gives richer outputs and is a good way to keep learning. If you want to try other scenarios, feel free to reach out to me.

8.FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to validate ticket creation, priority classification, Flow execution, urgent task creation, agent assignment, and Agentforce responses. Each of these features was verified to ensure correct functionality and performance.

ADVANTAGES & DISADVANTAGES

Advantages:

- Automatic Ticket Prioritization – Sorts support tickets into High, Medium, or Low based on the ticket description.
- Quicker Spotting of Critical Issues – Lets support agents concentrate on urgent tickets first.
- Less Manual Effort – Automates ticket analysis, priority setting, task creation, and assignment.
- Smart Ticket Handling – Agentforce reviews ticket details and starts the right backend automation.
- Automatic Urgent Tasks – High-priority tickets automatically get an Urgent Ticket Handling task.
- Better Support Assignment – Routes High-priority tickets to the suitable support level, such as a senior support agent.
- Conversational Support – Support users chat with Agentforce to analyze tickets and get the result.
- Faster Response – Automation helps reduce delays when handling critical customer issues and urgent requests.
- SLA Risk Awareness – Older unresolved tickets can be checked for possible SLA breach risk.

- Scalable Support Operations – Salesforce automation can cope with growing ticket volumes.
- Higher Team Productivity – Removes repetitive work so support teams can focus on resolving tickets.

Disadvantages

- Keyword-Based Priority Logic – The priority classification depends on configured keywords such as urgent, not working, failure, issue, slow, and delay, so wording matters.
- Limited Grasp of Context – The current setup may miss some urgency signals because the logic is based only on configured conditions and not on the wider context of each ticket description.
- Reliance on Accurate Ticket Data – Analysis only works when Account, Contact, and ticket information is correct.
- Flow Maintenance Needed – Any change to ticket handling means updating the Auto-Launched Flow and its logic.
- Agentforce Setup Dependency – Agentforce replies and actions rely on correctly configured subagent instructions and Flow actions.
- Narrow Agentforce Scope – The subagent covers ticket analysis, priority decisions, and backend automation only; unrelated requests such as billing or subscriptions are out of scope.
- Basic SLA Monitoring – The current build includes only an optional SLA-risk check on older tickets, not a comprehensive SLA monitoring and escalation solution for the whole support team.
- Assignment Logic May Grow – More support teams or advanced assignment rules would need extra configuration.
- Ongoing Upkeep – Priority keywords, Flow variables, Agentforce outputs, and assignment logic may need updates as requirements change.
- Limited Analytics for Now – Advanced analytics covering priority trends and

overall resolution performance are listed as future scope and are not part of what has been implemented in the current build.

9. Conclusion:

The Customer Support Ticket Priority Prediction and Automated Assignment System shows how Salesforce Flow and Agentforce can cooperate to lower manual ticket-handling effort. It fetches the latest ticket for an Account, analyzes the description, assigns a priority, creates an urgent task for High-priority cases, sets a support level, and replies conversationally. The result is quicker ticket handling, less manual work, smarter prioritization, and a more productive team.

10. FUTURE SCOPE

- Introduce additional priority indicators and business rules.
- Expand Agentforce outputs with more ticket details.
- Build more advanced SLA monitoring and escalation.
- Support more support teams and assignment rules.
- Add analytics on priority trends and resolution performance.
- Add further variables and map them in the Agentforce action so that responses carry richer output.

11. APPENDIX

- A. Custom Object: Support Ticket Intelligence
- B. Key Salesforce Records: Account, Contact, User, Task
- C. Flow: Support_Ticket_Intelligence – Auto-Launched Flow
- D. Agentforce Subagent: Support Ticket Priority Analysis
- E. Agentforce Action: Flow-based Support Ticket Intelligence action

Thank You

